

DSA 8420 Spring 2025

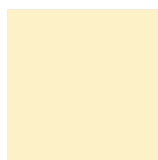
Homework 4

Due February 11, 2025

1. (10 points) Determine whether the following sets (colored areas) are convex. If they are, mark or describe the set of all extreme points.



(a)



(b)



(c)

2. (14 points) Given the following system of linear equations.

$$\begin{aligned}x_1 + 3x_2 - x_3 + x_4 &= 30 \\ 2x_1 + x_2 + 2x_3 + x_4 &= 15\end{aligned}$$

There are six basic solutions to the system. List any two of them. For each listed basic solution, specify the basic and nonbasic variables and the basis matrix B .

3. Consider the following polyhedron in standard form

$$P = \{ \mathbf{x} \in \mathbb{R}^5 \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \},$$

where

$$A = \begin{bmatrix} 2 & 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 7 \\ 2 \\ 3 \end{bmatrix}.$$

(a) (8 points) Find the basic solution $\mathbf{x} \in \mathbb{R}^5$ corresponding to the

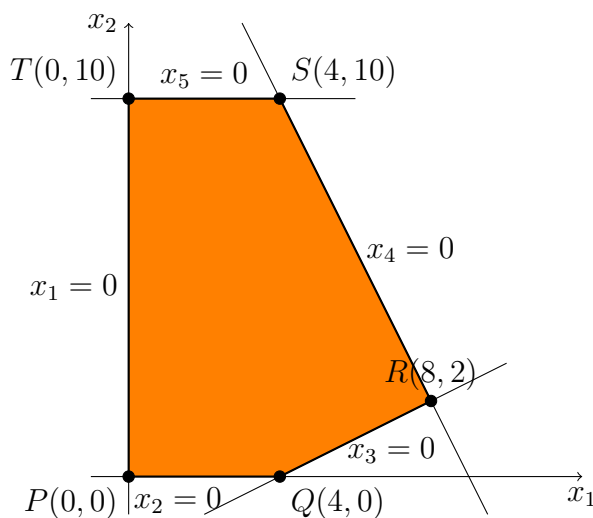
$$\text{basis matrix } B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

(b) (2 point) Is $\mathbf{x}$ a basic feasible solution?

4. Consider the following polyhedron in $\mathbb{R}^2$ defined by

$$\begin{cases} x_1 - 2x_2 \leq 4 \\ 2x_1 + x_2 \leq 18 \\ x_2 \leq 10 \\ x_1, x_2 \geq 0 \end{cases},$$

whose extreme points are marked on the graph.



By adding slack variables to the inequalities, we can transform the above inequality system to the following linear system with five variables:

$$S = \{ \mathbf{x} \in \mathbb{R}^5 \mid A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \},$$

where

$$A = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 4 \\ 18 \\ 10 \end{bmatrix}.$$

- (a) (8 points) Specify the basis matrix B corresponding to the extreme point $S(4, 10)$.
- (b) (8 points) Find a basis B such that the corresponding basic solution is not feasible.